

# Microeconomic Foundations, I: Choice and Competitive Markets

## Instructor's Manual

---

## Chapter 15: General Equilibrium, Efficiency, and the Core

### Notes on the Chapter

This chapter primarily concerns the First and Second Theorems of Welfare Economics, the extension of the First Theorem to the Core (every Walrasian equilibrium allocation is a core allocation), and the Debreu–Scarf Theorem. There is a bit of “conversation” following Debreu–Scarf concerning Aumann’s approach to the connection between Walrasian-equilibrium allocations and the core for “large” economies (that is, for a nonatomic economy, core allocations *are* Walrasian-equilibrium allocations). And, at the end, the “problem” of externalities is introduced, and the notion of a Lindahl equilibrium is provided.

This is all fairly straightforward and standard stuff, and I doubt you will find anything here that surprises you. Perhaps it is worth noting the following:

- I am a bit long-winded in the proof of the First Theorem: I describe what is going on as using equilibrium prices to separate the set of feasible bundles from the set of bundles that could be allocated in a Pareto-superior way to the equilibrium allocation. While this does make the proof lengthier than is required, it makes the transition to the Second Theorem very easy and intuitive.
- I spend a bit of space on various definitions of what a coalition might “control” when it comes to showing that Walrasian-equilibrium allocations are in the core of an economy with production.
- After giving the “real” proof of Debreu–Scarf (separation of the strict negative orthant from the convex hull of the union over consumers of utility improving net trades), I provide a somewhat informal proof that, in my experience, appeals more to the intuition of students.
- That said, just before the book was to be sent off to the printers, the issue of cover design came up, and I was asked to suggest a single “picture” that captured the essence of the book. The cover of the book was the result and, in retrospect, I wish

I had drawn this picture in the text, as it has helped me better to understand how the real proof works. I was able to insert a final problem into the text before it went to printers, Problem 15.12, I think this is a good problem to assign and to discuss before getting into the details of the proof of Debreu–Scarf.

Teaching this material is as straightforward as the material itself. Being a bit pressed for time, I have tended to gloss over the technicalities in the proof of the Second Theorem; if the students “get” the idea of the proof of the First Theorem, they can “see” how the Second Theorem will be proved, albeit without dealing with those technicalities. I have spent a fair bit of time on the “strategy” for proving Debreu–Scarf, because I find the basic idea of the proof so wonderful.

## Notes on the Problems

Problem 15.2 is, in my opinion, essential. Problem 15.12, discussed above, can be very helpful. Otherwise, this is a not-very-inspiring set of exercises, mostly simply drilling on technique. Note, however, that Problem 15.3 may lead to the question, “Are  $\mathcal{PS}(x)$  and  $PS(x)$  convex sets?” the answer to which I do not know. (See the discussion of this point following.) If you are going in for a discussion of externalities (and, perhaps, other ways that the Theorems of Welfare Economics might fail), an example is useful, and Problem 15.11 provides a simple one. (It also has the minor benefit of reinforcing material from Chapter 8 that students may have forgotten.)

## Solutions to Non-Starred Problems

■ 1. The change would not affect developments in any material way, but it would make life harder in the proof of the Second Theorem (Proposition 15.6). In the proof, we separate  $X^*$  from  $\mathcal{SPS}(x)$ , and announce that the separating price vector  $p$  is nonnegative, since  $X^*$  is comprehensive below. We could not longer make that claim directly, with the alternative definition. Indeed, suppose we defined  $X^* = \{x \in R_+^k : x \leq \sum_h e^h + \sum_f z^f, \text{ for some selection of production plans}\}$ . Consider a caricature economy with one consumer, two goods, and no firms, for which  $\sum_h e^h = (1, 1)$ . Suppose as well that the consumer’s preferences are given by  $u^h(x_1, x_2) = x_1 - x_2$ . This consumer is locally insatiable (and her preferences are continuous). There is a unique Pareto-efficient allocation for this economy, namely  $x$  in which  $x^h = (1, 0)$ . (Draw the picture or otherwise verify this!) And for this allocation,  $\mathcal{SPS}(x) = \{(x_1, x_2) : x_1 - x_2 > 1\}$ , while, with the amended definition,  $X^* = \{(x_1, x_2) : 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1\}$ . Amongst the many candidate separating hyperplanes are those with  $p_2 < 0$ , and we might even assert that the most natural separating hyperplane is  $p = (1, -1)$ , since this is proportional to the gradient of  $u^h$  at the Pareto-efficient point. But if we want a nonnegative price vector that separates the two sets, we must take  $(p, 0)$  for  $p > 0$ . If we define  $X^*$  as in the text, this is the only separating price vector. But with the alternative definition of  $X^*$ , you’d have to sweat a bit (not a lot) to be sure this is the one you get.

■ 3. The convenience of  $SPS(x)$  and  $\mathcal{SPS}(x)$  comes in showing that they are convex sets. Suppose  $\hat{x}$  and  $\check{x}$  are members of  $PS(x)$ . We know that there is some  $h$  such that  $\hat{x}^h \succ^h x^h$ , and some  $h'$  such that  $\check{x}^{h'} \succ^{h'} x^{h'}$ . But imagine that  $h \neq h'$ , and that  $\hat{x}^{h'} \sim^{h'} x^{h'}$  and  $\check{x}^h \sim^h x^h$ . When we take the convex combination  $\frac{1}{2}\hat{x} + \frac{1}{2}\check{x}$ , convexity of the consumers' preferences only tells us that  $\frac{1}{2}\hat{x}^h + \frac{1}{2}\check{x}^h \succeq^h x^h$ , and similarly for  $h'$ . We don't know that one or the other is strict preference, so we don't know that  $\frac{1}{2}\hat{x} + \frac{1}{2}\check{x} \in PS(x)$ . With strict Pareto superiority, this problem doesn't present itself.

That concerns "convenience." But it doesn't answer definitively the question, Are  $PS(x)$  and  $\mathcal{PS}(x)$  convex sets? Just because the simple proof fails doesn't mean that, perhaps, a more complex proof will show that these sets are indeed convex. Or, alternatively, perhaps a counterexample to convexity can be produced.

I don't know the answer to this question (and would be grateful if someone could supply either a proof or a counterexample; email me at [kreps@stanford.edu](mailto:kreps@stanford.edu)). Remember that we need preferences that are convex, continuous, and locally insatiable. Counterexamples are easy if we don't insist on local insatiability, but the combination of local insatiability and convexity has stopped me cold. Moreover (to give you a sense of where to look for a counterexample, if that is your wont), suppose  $\hat{x} \in PS(x)$  by virtue of consumer  $h$  (that is,  $\hat{x}^h \succ^h x^h$ ) and we know that  $\hat{x}^h$  is interior. With the additional assumption that  $\hat{x}^h$  is interior, a proof can be given: Suppose we are looking at the (strict) convex combination of  $\hat{x}$  and  $\check{x}$ , both from  $\mathcal{PS}(x)$ , in the mixture  $\alpha\hat{x} + (1 - \alpha)\check{x}$ . (We only have to worry about strict convex combinations, obviously.) Since  $\hat{x}^h \succ^h x^h$  and  $\hat{x}^h$  is interior, we can find an open neighborhood around  $\hat{x}^h$  in which strict preference over  $x^h$  continues to hold and all points in the neighborhood are nonnegative. By local insatiability, we can find some  $x \in R_+^k$  very, very close to  $\check{x}^h$  that is strictly preferred by  $h$  to  $\check{x}^h$  and hence strictly preferred to  $x^h$ . But  $\alpha\hat{x}^h + (1 - \alpha)\check{x}^h$  can be written as a convex combination of this  $x$  and a point in the neighborhood of  $\hat{x}^h$  that we described, as long as  $x$  is close enough to  $\check{x}^h$ . (The argument should be clear, so I won't spell out the details. Obviously, "close enough" will depend on  $\alpha$ .) But since all points in the neighborhood of  $\check{x}^h$  and this specific  $x$  are strictly preferred to  $x^h$ , so is this convex combination. This ensures that  $\alpha\hat{x} + (1 - \alpha)\check{x} \in PS(x)$ . In other words, if a counterexample can be produced, it will be necessary that any consumer who strictly prefers her part of  $\hat{x} \in PS(x)$  is being allocated a bundle  $\hat{x}^h$  on the boundary of her consumption set. Good luck on finding a counterexample or a proof; I could do neither.

■ 6. Suppose that, for some coalition  $J$ , we could find an allocation  $(x^j)_{j \in J}$  such that  $x^j \succeq^j x^j$  for all  $j \in J$ , and  $x^j \succ^j x^j$  for some such  $j$ . And suppose that for at least one of the  $j \in J$  for which  $x^j$  is strictly preferred to  $x^j$ ,  $x^j$  is strictly positive. Let us call this particular member of  $J$ ,  $j^0$ . And let  $L$  be the number of members of  $J$ .

By continuity of preferences, there is an  $\epsilon > 0$  such that (1)  $x_i^{j^0} > L\epsilon$  for  $i = 1, \dots, k$ , and (2) every bundle  $x$  that is within  $L\epsilon$  of  $x^{j^0}$  is strictly preferred to  $x^{j^0}$ .

By local insatiability, we can find for every  $j \in J$  that is not  $j^0$ , a bundle  $\hat{x}^j$  that (1) is within  $\epsilon$  of  $x^j$  and (2) is strictly preferred to  $x^j$ .

Let  $\hat{x}^{j^0}$  be defined as  $\hat{x}^{j^0} = x^{j^0} - \sum_{j \in J, j \neq j^0} (x^j - \hat{x}^j)$ . By the triangle inequality,  $\hat{x}^{j^0}$  is within  $L\epsilon$  of  $x^{j^0}$ , therefore it is strictly positive and is strictly preferred by  $j^0$  to  $x^{j^0}$ .

And  $\sum_j \hat{x}^j = [\sum_{j \in J, j \neq j^0} \hat{x}^j] + \hat{x}^{j^0} = [\sum_{j \in J, j \neq j^0} \hat{x}^j] + [x^{j^0} - \sum_{j \in J, j \neq j^0} (x^j - \hat{x}^j)] = \sum_{j \in J} x^j$ , therefore the allocation  $(\hat{x}^j)_{j \in J}$  is feasible for  $J$ . Finally, by construction,  $\hat{x}^j \succ^j x^j$  for all  $j \in J$ .

■ 8. Suppose that we have a Walrasian equilibrium with price vector  $p$ , at which two copies of consumer-type  $h$  choose distinct bundles, say  $x^1$  and  $x^2$ . Then both  $x^1$  and  $x^2$  are affordable for  $h$ ; and then so is  $1/2x^1 + 1/2x^2$ . And since preferences are strictly convex, this convex combination is strictly better than one of  $x^1$  or  $x^2$ , so one of the two (and perhaps both) is failing to maximize her utility.

■ 9. (I believe this is the third time you've been asked to prove this!) Suppose that  $x \geq x'$  and  $x \neq x'$ . If  $u$  is nondecreasing, then  $u(x) \geq u(x')$ . Since  $u$  is strictly quasi-concave,  $u(0.5x + 0.5x') > u(x')$ . But since  $x \geq 0.5x + 0.5x'$ ,  $u(x) \geq u(0.5x + 0.5x') > u(x')$ . Hence,  $u(x) > u(x')$ .

■ 10. The proof of Proposition 15.11 can be used, with only a few changes. Because preferences are not assumed to be convex, there is no guarantee that each  $Z^h(x^h, e^h)$  is a convex set. This doesn't affect the definition of  $Z^*(x, e)$  at all, but when it comes to constructing an element  $\zeta^0$  that is in  $Z^*(x, e)$  and in the strict negative orthant, you can't assume that  $\zeta^0$  is a convex combination of  $\zeta^h$  with at most one  $\zeta^h$  from each  $Z^h(x^h, e^h)$ . Instead,  $\zeta^0$  is a convex combination of elements from the  $Z^h(x^h, e^h)$ , where one or more of the  $Z^h(x^h, e^h)$  may contribute two or more terms to the convex combination. (By Caratheodory's Theorem, you know that no more than  $k + 1$  points are needed in the convex combination, but that is irrelevant to the current proof.) Therefore, when you go on to construct from the slightly perturbed  $\zeta^0$  (perturbed so that the scalars in the convex combination are of the form  $J/N$  for a fixed integer  $N$  and varying integers  $J$ ) the coalition and allocation that stands as an objection to  $x$ , you may need in your coalition representatives of a type  $h$  that are treated differently. But you still wind up in the same place: If  $x \in R_+^{kH}$ , when used to construct equal-treatment allocations for each replica economy  $\mathcal{E}^N$ , is in the core of each of those replica economies, then  $Z^*(x, e)$  must be disjoint from the strict negative orthant in  $R^k$ , and a separating hyperplane with  $p \geq 0$  and  $\beta \geq 0$  can be constructed.

The step where it is shown that  $p \cdot (x^h - e^h) \geq \beta$  needs modification: The argument used before relied on the assumption that each  $u^h$  is strictly increasing. But it is enough (for this step) that  $u^h$  is locally insatiable: The key to the argument is that we can find  $x^{h,\epsilon}$  that is strictly preferred to  $x^h$  and that is within  $\epsilon$  of  $x^h$ , and that is clearly an implication of local insatiability. Then  $x^{h,\epsilon} - e^h \in Z^h(x^h, e^h)$ , so  $p \cdot (x^{h,\epsilon} - e^h) \geq \beta$ , and now pass to the limit as  $\epsilon$  approaches zero.

Then the step that  $\beta = 0$  and  $p \cdot (x^h - e^h) = 0$  follows, and the demonstration that  $p$  together with an  $x$  forms a Walrasian quasi-equilibrium goes through without change.

But we can't show that  $p$  is strictly positive, and so we can't show that we have a (full-fledged) Walrasian equilibrium. Suppose, however, that consumer  $h$ 's allocation  $x^h$  is strictly positive. If  $\hat{x}^h \succ^h x^h$ , then by continuity of preferences,  $\alpha \hat{x}^h \succ^h x^h$  for some  $\alpha$  strictly less than (but presumably close to) 1. Since we have a quasi-equilibrium,  $p \cdot (\alpha \hat{x}^h) \geq p \cdot x^h$ , and since  $x^h$  is strictly positive and prices are nonnegative and nonzero,  $p \cdot x^h > 0$ . Therefore,  $p \cdot (\alpha \hat{x}^h) = \alpha(p \cdot \hat{x}^h) > 0$ , which implies  $p \cdot \hat{x}^h > \alpha(p \cdot \hat{x}^h) \geq p \cdot x^h$ . We have shown, then, that  $\hat{x}^h \succ^h x^h$  implies  $p \cdot \hat{x}^h > p \cdot x^h$ ;  $x^h$  is a full-fledged solution to consumer  $h$ 's utility-maximization problem, as long as  $x^h$  is strictly positive.